ATHARVA PARANJPE

Software Development Engineer II | Distributed Systems | AWS Infrastructure

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SUMMARY

Backend engineer with 6+ years building distributed systems, 5+ at AWS on Amazon Connect. Built the access-control layer and AgentCore/MCP infrastructure behind Connect Assistant, and shared platform services (feature gating, UI reverse proxies) used across Connect teams. Strong in Java, secure API design and production operations.

EXPERIENCE

Software Development Engineer II

Amazon Web Services – Amazon Connect

07/2021 – Present Seattle, WA

Connect Analytics 02/2025 – Present

- Designed and built the authorization layer for the analytics API behind Connect Assistant, authorizing each request with the caller's SigV4 credentials so results are scoped to exactly the data that caller may access
- Re-architected the API to serve multiple data sources (OpenSearch, a knowledge graph and a data lake) while preserving the same access controls
- Helped stand up the Amazon Bedrock AgentCore infrastructure that serves Connect Assistant's runtime behind the Connect console chat widget, backed by MCP servers built on that access-controlled analytics API
- Built scope reduction for the Connect Assistant agent by intersecting the agent's auth-token permissions with the invoking user's, so the agent can never access more than that user can
- Designed the data flow for tagging use cases, adding anti-entropy reconciliation to detect and repair drift and keep tag data consistent

Connect Resource Configuration 07/2021 – 01/2025

- Built a centralized feature-gating service with HTTP and Coral RPC APIs for frontend and backend callers; its decoupled deployment model removed a shared-pipeline bottleneck across multiple teams
- Set up the reverse-proxy stacks that front the entire Amazon Connect UI
- Helped remove a latency hop from UI loads by fetching static-asset manifests directly from S3 instead of through a versioning service layer, cutting P99 latency by 300 ms
- Owned the Routing Profile enhancement end to end with CloudTrail audit logging, and designed the CRUDL APIs for agent proficiencies that route contacts to the right agents
- Led the launch's operational readiness review and worked with security engineers and penetration testers to identify CVE risks

Grader and Part-Time Lecturer

Rutgers University

09/2019 – 05/2021 New Brunswick, NJ

- Part-time lecturer for CS170 (Foundations of Web Development) for two semesters; also graded CS512 (Data Structures and Algorithms) and was a research assistant at the School of Social Work

Technical Analyst

Credit Suisse

07/2018 – 06/2019 Pune, India

- Migrated the legacy ColdFusion watchlist page to AngularJS backed by Java RESTful services; relationship managers used it to suggest products to high-net-worth clients

SKILLS

LANGUAGES

Java | Kotlin | TypeScript | JavaScript | Python | SQL

AI & AGENTS

Amazon Bedrock | Bedrock AgentCore | Agentic AI | MCP | NLP / BERT | TensorFlow

AWS

IAM | SigV4 | Lambda | ECS | EC2 | Route 53 | API Gateway | DynamoDB | OpenSearch | Neptune | S3 | CloudFront | CloudTrail / CloudWatch | SQS / SNS | Kinesis | CDK / CloudFormation

FRAMEWORKS & TOOLS

REST APIs | Spring Boot | Coral RPC | Docker | React

SYSTEMS

CI/CD | Anti-entropy | Reverse proxies | Data lakes

EDUCATION

M.S. in Computer Science

Rutgers University

09/2019 – 05/2021

Machine Learning specialization · GPA 3.9/4.0

B.Tech. in Electronics and Telecommunication

VJTI, Mumbai

2014 – 2018

GPA 9.13/10

PUBLICATIONS

Facts That Matter

Poster paper, ISWC 2020

BERT-based neural model that ranks predicate-object pairs for natural-language queries, evaluated with NDCG

ACHIEVEMENTS

- **Winner, Sardar Patel Robotics Challenge**
  Inter-college robotics competition, 2015
- **Joint Runner-Up, Android App Hackathon**
  Annual college festival, 2017